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Government intervention, efficiency wages, and the employer size wage effect in Zimbabwe [1]

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Abstract

This paper uses matched employer–employee data from a survey of 201 manufacturing firms and 1609 of their workers conducted in Zimbabwe in the summer of 1993. The results indicate that there is a substantial premium associated with employment in larger firms, and that this premium cannot be explained by differences in worker quality and job characteristics, nor is it eliminated by controlling for unionization, minimum wages or other forms of government intervention. The size premium is much larger for white collar than for blue collar workers. These differentials are also found to be substantially larger than those estimated for other developed and developing countries. The analysis uses the data about firm characteristics to explore a number of efficiency wage-based explanations of the size differential, and finds results which are consistent with, but not conclusive proof of, hiring, turnover, and morale based notions of efficiency wages. © 1997 Elsevier Science B.V.

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1. Introduction

There is a large and growing empirical literature presenting evidence of the existence of wage differentials associated with the industry and size of firm where an individual is employed. These papers (Krueger and Summers, 1986, 1988, Brown and Medoff, 1989) argue that because these differentials cannot be fully explained by differences in worker quality or job characteristics affecting worker utility, they provide strong evidence against the hypothesis that labor markets are perfectly competitive. As an alternative, the literature explores the class of efficiency wage hypotheses which are based on the notion that firms find it profitable to pay wages that are higher than the market clearing level. The efficiency wage literature includes a number of different explanations for a positive relationship between wages paid and profits, and though the empirical literature generally does not attempt to distinguish among them, the broad conclusion is that, as a class, these models have some explanatory power.

This paper uses data from a 1993 survey of manufacturing firms and their employees in Zimbabwe to extend the literature on the employer size wage effect in two dimensions. First, the analysis allows a comparison of the magnitude of the size effect in Zimbabwe with the results of other developing and developed countries. A comparison of Zimbabwe with other countries may provide an insight into the relationship between overall labor market competitiveness, government intervention, and the size of this effect.

Second, the analysis uses matched employer–employee data from a sample of firms and their workers. Most of the empirical work in this area uses either individual surveys, which contain very limited information about the firm (generally only size class and industry), or employer surveys in which wage and human capital measures are labor force aggregates or averages. The dataset used in this analysis includes both detailed individual level data about wages and human capital and a great variety of information about the firms. This allows more direct examination of some of the central hypotheses about the causes of these differentials than is possible with either individual or firm level data alone. In particular, the rich set of information about firm behavior will allow us to examine empirically several major efficiency wage hypotheses. The specific institutional characteristics of Zimbabwe's labor markets are also important to this analysis.

The findings in this paper indicate that, controlling for human capital and job characteristics, there is a very large premium associated with working for larger firms in the Zimbabwean manufacturing sector, and that this effect is much stronger for white collar than for blue collar workers. This premium cannot be explained by a number of institutional features of the Zimbabwean labor market, including unionization, minimum wages, and government intervention of other types. This implies that larger firms must find it profitable to pay higher wages, as would be the case under a number of constructions of the efficiency wage hypothesis. The results show evidence supporting the idea that larger firms may use higher wages to increase the quality of their applicant pools, to reduce employee turnover, and to enhance worker loyalty. These hypotheses are all consistent with the larger premium for white collar workers, for whom these concerns may be more relevant.

The paper is organized as follows. Section 2 describes the dataset and presents some baseline estimates of the size effect. The hypotheses about the size effect based on worker quality and job characteristics are explored in Section 3. Institutional and government policy considerations are examined in Section 4. Section 5 describes a number of variants of the efficiency wage hypothesis, relates them to the size premium, and presents some empirical evidence allowing us to assess the relevance of each for the Zimbabwean data. Finally, some concluding remarks are presented and the implications of the results discussed in Section 6.

2. Data and variables

The data used in this study are drawn from a survey of 201 manufacturing firms in Zimbabwe conducted in the summer of 1993. The survey is part of the Regional Program on Enterprise Development (RPED), a multi-year study of the manufacturing sector in several African countries (Cameroon, Côte d'Ivoire, Ghana, Kenya, Rwanda, Burundi, Tanzania, Zambia and Zimbabwe). The RPED is organized by the Africa Region Technical Department of the World Bank, and funded by the Canadian and a number of European governments. The fieldwork for the Zimbabwe study was conducted by a team of researchers, including the author, from the Economic and Social Institute of the Free University of Amsterdam and the Department of Economics at the University of Zimbabwe.

The RPED is designed to provide an overview of the performance of manufacturing firms in the post-structural adjustment period, and focuses on a wide variety of aspects of firm behavior. The survey instrument includes modules covering: the history of the firm and the entrepreneur; current production, sales, capital stock, and costs; employment, labor force structure, compensation, benefits, and turnover; acquisition, adaptation and use of technology; firm response to the regulatory and infrastructure environment; and experiences with conflict and conflict resolution.

The sample frame was constructed from two sources. The Central Statistical Office (CSO) provided a list of all registered firms. Although firms of any size may be registered, it seemed unlikely that all of the small-scale enterprises would have done so. Therefore, the CSO list was used for firms with 50 or more employees and another source was used as a sampling frame for the smaller firms. It seems likely that the CSO list of firms with 50 or more workers provides virtually complete coverage of this population. The sampling frame for firms with fewer than 50 workers was drawn from the GEMINI survey conducted in August 1991 by Mead and others from Michigan State University. This was a stratified cluster sample of households in eight areas in Zimbabwe in which interviews at households and shops were used to locate enterprises employing 50 or fewer people and marketing at least half of their output. The survey generated a list of 5575 enterprises, of which 107 were manufacturing firms with at least five workers and included in the RPED sampling frame. [2]

The two lists were combined to create the sampling frame. Sampling was done on the basis of firm size, so that every worker in the sample frame had an equal probability of being drawn. In essence, a worker was drawn and the corresponding firm was included in the sample. Unlike a sample based on enterprises, workers in small firms would not be systematically over-sampled by this technique.

The instrument also includes a survey for a sample of each firm's employees. A maximum of 10 workers were interviewed in each firm (less than 10 interviews in firms with fewer than 10 employees). The fact that fewer workers were interviewed in small relative to larger firms implies that workers in small firms are undersampled in this analysis. Each firm was asked to indicate how many workers it employed in each of 10 broad occupational categories: management, administration and clerical, technical (primarily engineers and scientists), sales, equipment maintenance and repair, factory supervisors and foremen, skilled production workers, semi-skilled and unskilled production, support staff (janitors, night watchmen, canteen staff), and apprentices. The sample was drawn according to these occupational categories in proportion to their representation in the firm's total labor force. During the workers' survey, individuals were asked to describe their jobs, and assigned an occupational category by the enumerator. These categories are so broad and generic that it seems unlikely that job classifications were influenced by firm size or jointly endogenous with wages. The survey questions cover current wages, occupation, union status, education, tenure, apprenticeship history, layoff experience, as well as some very limited demographic information. The 201 firms in the study yielded a sample of 1717 workers. Missing data needed for the testing of some hypotheses resulted in the elimination of 108 observations, [3] leaving a sample of 1609 workers to be used for the analysis in this paper. The analysis will use data from both the firm and worker surveys.

The sample was drawn from four manufacturing sectors: food and beverage processing, textiles and garments, woodworking and furniture making, and metal working. These four sectors account for 75% of manufacturing employment in Zimbabwe. The final sample of 1609 workers includes 392 (24%) in the food and beverage sector, 202 (13%) in wood and furniture, 712 (44%) in textiles and garments, and 303 (19%) in metal working. This distribution is quite similar to the distribution of workers across these sectors in the labor force as a whole.

[2] Details about the GEMINI survey can be found in Mead et al., 1993.

[3] Of these 108, 69 were eliminated because the year the individual finished school was missing, 21 because of missing levels of schooling, 22 because of missing current wages, and 5 because of missing tenure at the current employer. There was some overlap in the missing data.

A complete listing of the variables used in the paper is provided in Appendix A. Throughout the analysis, the dependent variable is log hourly earnings, which is weekly earnings divided by usual weekly hours. Earnings include both wages and the value of cash and in-kind allowances paid by the employer. These allowances include employer-provided or subsidized meals and protective clothing, cash and in-kind payments for housing and transportation, and any bonuses, including the Christmas bonus. These allowances are an important part of compensation, particularly for white collar workers and in larger firms. The wage and hours data are drawn from the workers' own reports, while the allowances are as reported by the firm, since the workers' survey did not ask about allowances. For each worker, the average allowances the firm reported paying to workers in his occupational category were added to reported wages to produce earnings.

One clear advantage of this dataset is that it contains precise information about firm size as measured by total employment, rather than simply the size category of a firm in which the individual is employed. The size variable, reported by the firm, is firm size rather than establishment size, as the unit of observation in the survey was the enterprise. These data permit estimation of the employer size effect using a continuous size variable. Throughout the analysis, the continuous size measure used is ln(size), which was selected both because it is standard in studies using continuous measures (see Brown and Medoff, 1989), and because the relationship between earnings and firm size appears to be nonlinear, and the log measure captures some of this nonlinearity. The magnitude of the size effect can be measured as the percentage increment associated with working for a firm whose ln(size) is one standard deviation above the mean relative to one whose ln(size) is one standard deviation below the mean. If we average across the workers in the

Table 1
Summary statistics by firm size

	All	Small	Medium	Large	Very Large
Number of Workers	1609	149	556	456	448
% of Total	100	9.3	34.6	28.3	27.8
Mean Wage	5.86 (8.30)	2.06 (1.56)	4.44 (5.78)	5.95 (7.73)	8.80 (11.35)
Mean Earnings	7.08 (10.14)	2.38 (1.56)	5.35 (7.40)	7.15 (9.53)	10.72 (13.37)
Mean Experience	18.55 (11.25)	13.15 (10.78)	18.96 (11.90)	20.02 (10.78)	18.35 (10.50)
Mean Grade Completed	8.69 (2.94)	8.95 (2.70)	8.32 (2.80)	8.56 (2.93)	9.19 (8.13)
Mean Tenure	9.58 (8.06)	3.95 (4.56)	8.85 (7.66)	11.02 (8.26)	10.89 (8.09)
Earnings Distribution					
90%	16.40	4.07	11.25	16.05	26.43
75%	6.37	2.75	4.84	6.60	12.30
Median	3.37	2.02	2.79	3.56	4.88
25%	2.44	1.45	2.27	2.72	3.08
10%	1.94	0.96	1.78	2.22	2.52

Table 2
Baseline size coefficients

Specification	Controls	Size Measure	Coefficient (t-statistics)	Percent Premium	R^2
Whole Sample ($N = 1609$)					
(1)	None	ln(size)	0.257 (12.9)	79	0.205
(2)	None	Medium	0.571 (5.07)	77	0.153
		Large	0.854 (8.11)	135	
		Very Large	1.19 (10.7)	228	
White Collar ($N = 552$)					
(1)	None	ln(size)	0.311 (11.2)	93	0.239
(2)	None	Medium	1.03 (5.06)	180	0.192
		Large	1.28 (6.65)	260	
		Very Large	2.72 (8.91)	450	
Blue Collar ($N = 1057$)					
(1)	None	ln(size)	0.153 (8.28)	47	0.183
(2)	None	Medium	0.376 (3.64)	46	0.133
		Large	0.507 (5.29)	66	
		Very Large	0.705 (6.69)	102	

sample, we find that the mean firm size is 335, the median is 125, and the standard deviation is 676. The mean and standard deviation of ln(size) are 4.7 and 1.55 respectively.

For purposes of comparison with studies using discrete size measures, and to facilitate the presentation of summary statistics throughout the paper, the analysis also uses a set of four discrete size categories. Small firms have 1 to 10 workers, medium 11 to 100, large 101 to 250, and very large firms have more than 250 workers. At some points in the analysis other discrete measures, comparable to those used in other papers, will be used. In all the regression analyses the size effects with these discrete measures are relative to employment in small firms.

Summary statistics by firm size class are presented in Table 1. Table 2 presents baseline size coefficients from a regression of earnings on firm size including no control variables. Estimates are presented for the whole sample as well as separately for white and blue collar workers. [4] Both tables show a clear pattern of wages rising dramatically with firm size for all groups of workers, and a

[4] Occupation-specific continuous size coefficients ranged from 0.088 for apprentices to 0.369 for managers. F-tests of differences in these coefficients across occupations allowed the sample to be divided into the white and blue collar groups. White collar workers include management, administration, commercial, technicians, supervisors and equipment maintenance staff. Blue collar includes skilled and unskilled production workers, apprentices, and support staff.

substantially larger effect for the white collar workers. The remainder of this analysis will present and test a number of hypotheses about the source of this differential and the variation in its magnitude between the occupational groups.

3. Market clearing hypotheses

3.1. Labor quality differences

The basic labor quality difference argument is that larger firms hire 'better' workers, and therefore pay higher wages. The reasons for this preference can be most easily understood with reference to the greater capital intensity of larger firms. To the extent that physical and human capital are complements, firms using more physical capital will also require more human capital, hiring workers with larger human capital endowments, and higher productivity, and paying higher wages.

If this hypothesis is true, then the employer size wage effect should be explained, and hence the significant positive coefficient on firm size eliminated, by controlling for worker quality in the estimated earnings functions. Regressions including controls for measurable human capital are presented in columns 1 and 2 of Table 3. [5] While the effects of firm size are smaller than those presented in Table 1, they are certainly not eliminated. [6] Inclusion of these controls has similar effects on the size coefficients for the white and blue collar workers separately.

It is possible that workers in larger firms are not only of better quality when hired, but that they acquire additional human capital on the job. If larger firms provide more firm-specific human capital than do smaller ones, perhaps because of their use of more sophisticated technologies, this would result in different earnings profiles for workers in larger firms. If the workers share in the investment in specific training, they would receive wages lower than they could get elsewhere during the training period, and higher wages afterward. Since workers with specific human capital are less likely to quit their jobs, average wages in larger firms would appear higher.

The ideal means of testing this hypothesis would be to identify workers who are currently receiving training and those whose training is completed, and include these variables in the earnings function. We would expect a negative coefficient on

[5] In these and all other regressions in the paper, standard errors have been corrected for heteroskedasticity using a version of the method of White (1980) appropriate for grouped data.

[6] This analysis uses potential experience, or the number of years since the individual left school, as its measure of experience. An alternative specification using the worker's age and its square yields virtually identical size coefficients, so the large size of these coefficients relative to those seen in other countries is probably not an artifact of differences in the appropriateness of the potential experience measure.

the current training variable and a positive coefficient on completed training, and we would expect the size effect to be reduced. Unfortunately, these data only contain information about current on and off the job training, and it is not obvious what results that should produce. Columns 3 and 4 in Table 3 present estimates including these controls.

The coefficients on the training variables are significant and positive, and the size coefficients are reduced but not eliminated by their inclusion. This result is not consistent with the specific training story told above, but there are other plausible interpretations. Some firms may both pay premium wages and provide more training, perhaps as a means of reducing worker turnover, as will be discussed later. Alternatively, workers in all firms may be positively selected into training. In that case, the training variable would be a proxy for other attributes of worker quality, and would be accounting for the better quality of workers in larger firms.

The labor quality controls necessarily omit unmeasured dimensions of quality. To the extent that these characteristics are correlated with the measured character-

Table 3
Earnings regressions including size and labor quality controls, dependent variable is log hourly earnings, absolute value of t-statistics in parentheses, $N = 1609$

	(1)	(2)	(3)	(4)
Constant	−0.818 (5.93)	−0.629 (4.30)	−0.807 (5.97)	−0.650 (4.51)
Experience	0.057 (11.4)	0.059 (11.8)	0.060 (12.0)	0.063 (12.6)
Experience^2/100	−0.089 (8.90)	−0.095 (8.63)	−0.092 (9.30)	−0.097 (9.70)
Tenure	0.005 (1.67)	0.007 (2.33)	0.005 (1.67)	0.006 (2.00)
Primary	0.197 (2.05)	0.243 (2.47)	0.191 (2.05)	0.230 (2.39)
Some Secondary	0.663 (6.37)	0.727 (6.79)	0.645 (6.32)	0.702 (6.68)
Complete Secondary	1.079 (8.23)	1.169 (8.47)	1.010 (7.65)	1.085 (7.80
University	0.730 (2.91)	0.798 (3.59)	0.790 (3.28)	0.859 (3.95)
Vocational	−0.143 (1.10)	−0.142 (1.04)	−0.112 (0.88)	−0.109 (0.83)
Polytechnical	0.519 (2.91)	0.561 (3.04)	0.524 (2.72)	0.566 (2.94)
Former Apprentice	0.564 (7.20)	0.572 (6.81)	0.509 (6.44)	0.512 (6.02)
On the job Training	–	–	0.208 (2.57)	0.226 (2.67)
Outside Training	–	–	0.258 (4.22)	0.283 (4.71)
Male	0.076 (1.46)	0.095 (1.82)	0.073 (1.43)	0.091 (1.75)
European	1.251 (11.2)	1.280 (11.4)	1.276 (11.2)	1.301 (11.4)
Asian	0.687 (3.25)	0.663 (3.29)	0.664 (3.06)	0.639 (3.06)
Harare	0.161 (2.15)	0.199 (2.31)	0.174 (2.42)	0.207 (2.55)
Bulawayo	0.002 (0.26)	0.031 (0.37)	0.020 (0.27)	0.045 (0.57)
Ln(Size)	0.178 (9.88)	–	0.159 (8.83)	–
Medium	–	0.347 (3.85)	–	0.329 (3.73)
Large	–	0.535 (5.88)	–	0.484 (5.56)
Very Large	–	0.804 (8.20)	–	0.721 (7.36)
Adjusted R^2	0.490	0.472	0.504	0.489

istics included as controls, this will influence the coefficients on the human capital variables, but should not result in the presence of a significant size effect. If, however, the unmeasured aspects of worker quality are correlated with size but not with the measured characteristics, they could generate a size effect even when controls for measurable quality differences are included. Unfortunately, the data available do not allow us to control for these unobservables in any meaningful way. [7] It seems unlikely, however, that unobservable labor quality differences are the primary source of the remaining size differential. Given that inclusion of observable ability measures only reduces the magnitude of the size coefficient by one-third, and given the weak relationship between education and firm size shown in Table 1, it does not seem likely that controlling for unobservable labor quality would reduce the size coefficient to zero. If the training measures discussed above are indeed capturing unobserved worker quality, then the results in Table 3, indicating that their inclusion generates only a small reduction in the size coefficient, further support this view. There does seem to be a wage differential requiring an explanation other than differences in worker attributes.

3.2. Comparison with other results

The size effect estimates from Table 3 are larger than those reported in the literature. Brown and Medoff (1989), using continuous size measures from a number of U.S. datasets, report two standard deviation size wage premiums, controlling for worker quality, of 6% to 15%. Idson and Feaster (1990), using CPS data classified into five size categories (1–24, 25–99, 100–499, 500–999, and 1000 workers and more), found size effects for the four largest size categories relative to the smallest of 7.9%, 16.4%, 19.8% and 23.8% respectively. Using these size categories with the data used for this paper yields size effects of 27%, 51%, 109%, and 110% for these same size classes.

The Zimbabwean size effects are closer in magnitude to, but still larger than, those found by other authors using datasets from developing countries. Schaffner (1994), using data from the Peruvian LSMS in four size categories (2–5, 6–20, 21–200, and 200 workers and more) found size effects for the three largest categories of 27%, 49% and 73% respectively. Using these size categories with the data for this paper yields size effects of 75%, 117% and 192%. Little et al. (1987), using data for 5000 men in India, found that workers in firms with more than 500

[7] Attempts to use selection correction techniques to account for the unobservables have produced generally implausible results, which are available from the author on request. This is not surprising given that Schaffner (1994), using a dataset richer in individual demographic and family background information, finds that all such results are highly sensitive to changes in identifying restrictions and functional form, and do not provide trustworthy evidence about selection effects.

workers earn 42% more than those in firms with 10 to 99 workers. A similar comparison with Zimbabwean data yields a size effect of 68%. [8]

If these premiums are indeed measures of the deviation of labor market outcomes from those predicted by perfect competition, then it may not be surprising that these effects are larger in Zimbabwe than elsewhere. African labor markets have long been characterized by extensive government intervention acting to blunt the influence of market forces, and the types of information and monitoring problems that give rise to efficiency wages may be more severe in Zimbabwe than they are in more developed economies. Subsequent sections of this paper will address these issues more directly.

3.3. Job quality and compensating differentials

The principle of compensating differentials implies that workers who face less attractive working conditions will receive higher wages in compensation, even when the quality of these workers is the same as the quality of those working in better conditions. In order for these differences to explain the size wage differential, jobs in larger firms must have more unattractive characteristics than jobs in smaller firms.

Working conditions include both physical characteristics regarding the difficulty, danger and unpleasantness of the work, as well as more subjective characteristics of the work atmosphere such as relationships in the workplace and the tedium of the tasks. The RPED data include some items which can be used as proxies for working conditions, and this section will use them to test the hypothesis that the size premium can be explained by compensating differentials.

A first step is to include controls for industry and occupation. Working conditions probably differ across industries and industry composition does differ by firm size in this sample, with food and textile firms being generally larger than those in the wood and metal industries. Similarly, occupational differences also may engender differences in working conditions, both physical and in terms of working atmosphere. The data include only four industries and ten occupational characteristics, but Brown and Medoff (1989) found that including more detailed industry and occupation dummies has no effect on the size coefficient. Estimates of the size coefficients including these controls are presented in Table 4. The industry controls include three dummies (textiles is the omitted category) and occupation controls include 9 dummies (unskilled production workers is the reference group). Columns 1 and 5 present, for each specification, the estimates without occupation and industry controls from Table 3.

[8] All of these datasets include workers from a number of sectors other than manufacturing. This may generate a bias in the comparison, although the direction of that bias is not obvious. The data of Schaffner (1994) indicate that, in Peru, the size dispersion in wages is smaller in manufacturing, agriculture, construction and services than it is in other sectors. If that is the case elsewhere, then these comparisons understate the difference in the size premium between Zimbabwe and the other countries.

Table 4
Earnings regressions including industry and occupation controls, dependent variable is log hourly earnings, absolute value of t-statistics in parentheses, $N = 1609$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ln(Size)	0.159 (8.83)	0.161 (9.65)	0.158 (9.08)	0.164 (0.977)	–	–	–	–
Medium	–	–	–	–	0.329 (3.73)	0.322 (4.09)	0.380 (3.99)	0.367 (4.19)
Large	–	–	–	–	0.484 (5.56)	0.502 (6.33)	0.504 (5.31)	0.517 (5.89)
Very Large	–	–	–	–	0.721 (7.36)	0.750 (8.31)	0.732 (6.99)	0.769 (7.85)
Industry Controls	no	yes	no	yes	no	yes	no	yes
Occupation Controls	no	no	yes	yes	no	no	yes	yes
Adjusted R^2	0.504	0.526	0.628	0.652	0.489	0.515	0.613	0.641

All regressions include controls for experience and its square, tenure, schooling as in Table 3, training, race, gender and location.

Inclusion of these controls actually increases the magnitude of the size coefficients in both specifications, indicating that even if industry and occupational differences are associated with differences in working conditions and earnings, they are not a source of compensating differentials which reduce the firm size earnings premium. [9] This result holds for the occupational sub-samples as well.

The data do allow measurement of, or proxies for, more specific aspects of working conditions. In order for these differences to reduce the firm size effect, working conditions must deteriorate as firm size increases. As we shall see, while addition of these controls does reduce the size coefficients, it is also generally the case that better working conditions are associated with larger employer size.

One possible source of working conditions variation is working hours. If longer hours of work are a source of disutility for workers, one might expect those working longer hours to receive higher wages. Mean usual weekly hours as reported by the worker do vary with firm size. However, longer hours are associated with employment in smaller firms. If we restrict the sample to full time workers only (full time being more than 35 hours per week), the relationship is stronger. [10]

While the data do not include measures of how hazardous the work is, they do include some measures which can be considered as proxies for aspects of the physical comfort of the workplace. The firm data include information about use of utilities, allowing construction of two dummy variables. NOELEC = 1 if the firm has no electricity, while NOPHONE = 1 if the firm has no telephone. The electricity measure is obviously capturing the use of mechanized equipment, but it also proxies for the availability of lighting, cooling or heating of the premises, and the presence of indoor plumbing. Both measures can also be considered as proxies for working in 'modern' facilities. Not surprisingly, the fraction of workers in facilities without phone or electricity decreases sharply as firm size rises. [11]

A third commonly mentioned source of compensating differentials is commuting costs. If larger firms employ workers from a larger geographical area, these

[9] The results do indicate that earnings differ across sectors, with workers in the metal industry receiving earnings approximately 25% higher than those in the other sectors.

[10] Because the hourly wages are calculated as weekly wages divided by weekly hours, the inclusion of hours on the right hand side introduces division bias, which implies that the effect of hours on wages is probably overstated, although it is clear that average work weeks in smaller firms are significantly longer than in larger ones.

[11] Another source of differences in working conditions is the need to work evening or night shifts, the inconvenience of which may generate compensating differentials. Because the survey was conducted during daytime hours, none of the workers surveyed are night shift workers. However, the firm was asked if it generally conducted multiple shifts, and a dummy variable indicating that it does so was generated. At best, this variable proxies for the possibility that workers might be asked, or forced, to work nights. It also, however, probably proxies for capital intensity as well. Working in a firm with multiple shifts is positively correlated with firm size and a significant positive influence on wages. Inclusion of this measure in earnings functions substantially reduces measured size coefficients.

Table 5
Earnings regressions including working conditions controls, dependent variable is log hourly earnings, absolute value of t-statistics in parentheses, $N = 1609$

	(1)	(2)	(3)	(4)
Ln(Size)	0.159 (8.83)	0.141 (7.19)	–	–
Medium	–	–	0.329 (3.73)	0.248 (2.71)
Large	–	–	0.484 (5.56)	0.380 (4.02)
Very Large	–	–	0.721 (7.36)	0.604 (5.80)
Hours	–	−0.017 (5.42)	–	−0.019 (5.77)
No Electricity	–	−0.411 (2.45)	–	−0.382 (2.22)
No Telephone	–	0.014 (0.12)	–	−0.047 (0.34)
Bad Transport	–	0.019 (0.35)	–	0.048 (0.84)
Adjusted R^2	0.504	0.520	0.489	0.509

All regressions include controls for experience and its square, tenure, schooling as in Table 3, training, race, gender and location.

workers will have greater time and/or money costs of commuting to work, and should receive higher wages in compensation. Many of the firms and workers in Zimbabwe complained that traffic congestion and inadequate bus services generated extremely long commutes. Unfortunately, the data do not include individual measures of commuting time or costs. They do, however, include a series of questions about the firm's opinion of the quality of infrastructure. Owners and managers were asked to rank the problems posed by transport for workers on a scale from 1 to 5, where 1 signifies that transport posed no problem. From this data, we define a dummy variable indicating that the firm felt that worker transport was a problem, rating it 2 or greater. [12] The firm's perception that transport is a problem may proxy for the distance and/or time that its labor force commutes, and therefore for the firm's need to pay compensation. Larger firms are indeed more likely to consider worker transport a problem.

Table 5 presents the effects of including these measures of working conditions in the earnings regressions. Again, columns 1 and 4 reproduce results from Table 3 as a reference. The inclusion of the physical measures does reduce the size coefficients. The coefficients on the hours and electricity variables, however, indicate that 'worse' working conditions are associated with lower rather than higher wages, while the effects of having no telephone and bad transport have the expected sign. [13] This gives rise to the possibility that, rather than controlling for the existence of compensating differentials, these measures are picking up a distinction between some 'bad' jobs, characterized by long hours, unpleasant

[12] An alternative specification, in which the dummy is based on an employer rating of 3 or greater, yields qualitatively similar results.

[13] If we omit those variables which take the 'wrong' sign, the measured size coefficient is 0.148, which is larger than that found when these factors are included.

Table 6
Tenure and quit rate regressions, absolute value of t-statistics in parentheses

Dependent Variable	Controls	Size Measure	Coefficient
Ln(Tenure)	Union Membership, Race, Gender, Education, Training, Experience Location, Industry, Occupation	ln(size)	0.150 (4.775)
Ln(Tenure)	Union Membership, Race, Gender, Education, Training, Experience Location, Industry, Occupation	Medium	0.360 (1.785)
		Large	0.650 (3.281)
		Very Large	0.695 (3.226)
Ln(quitrate/(1-quitrate))	Unionization, share of production workers, share of male workers, ln(average hourly earnings)	ln(size)	−0.328 (3.293)
Ln(quitrate/(1-quitrate))	Unionization, share of production workers, share of male workrs, ln(average hourly earnings)	Medium	−0.85 (2.009)
		Large	−1.39 (2.729)
		Very Large	−1.44 (2.764)

physical conditions, and low wages, and 'good' jobs with the opposite characteristics. This would be consistent with a segmented labor markets approach, in which a secondary labor market characterized by low wages and poor working conditions exists alongside the primary labor market, in which one might expect working conditions to be negatively correlated with wages. Regardless of the interpretation one takes, these controls certainly do not eliminate the differential associated with working for a large firm, either in the whole sample or in the two occupational sub-samples. Better controls for job quality could conceivably eliminate the size effect, though that seems unlikely given how large the effect continues to be relative to those seen elsewhere.

The hypotheses about differences in labor quality and job quality can also be examined using data on quit rates and tenure in the firm. If the labor quality hypothesis is true, then workers in larger firms would be receiving the same wage they would receive elsewhere, and may be no less inclined to quit jobs in large firms than they are to quit those in small firms. This implies that, holding earnings constant, firm level quit rates should be unrelated to firm size, as should individual tenure with a given employer.

If the higher wages paid by larger firms perfectly compensate for inferior working conditions in those firms, then workers would be indifferent among jobs in different size firms. Under these strong assumptions, once we control for earnings, we would expect to see no correlation between firm size and quit rates or tenure. If the higher wages paid in larger firms actually constitute a wage premium, we would expect to see a negative correlation between firm size and quit rates. Estimates of these relationships are presented in Table 6. The top two rows examine the tenure effects, while the second two use data from 185 firms (missing data caused 16 firms to be dropped) to examine the effects of size on quit rates. The results of both tests contradict the implications of the compensating differentials hypothesis, as quit rates decrease and tenure increases as firm size rises. They are consistent with the labor quality hypothesis only if the relevant difference in labor quality across firm sizes is that larger firms choose workers who are less likely to leave their jobs.

These results suggest that the earnings differential associated with employment in larger firms is indeed a premium, in the sense that it cannot be explained fully by market clearing explanations consistent with perfect competition in labor markets. The remainder of this analysis will examine alternative explanations for the existence of this premium.

4. Labor market institutions and government intervention

4.1. Background

In the period between Zimbabwe's independence (1980) and the introduction of the Structural Adjustment Program in 1990, the government engaged in extensive labor market interventions, getting involved in hiring, wage setting, layoffs and dismissals. As Shadur (1994) points out, during this period the government actually took the role of labor's advocate, and created tripartite institutions composed of labor, management and the government to enforce labor regulations. Changes in statutes in 1990 brought about the abolition of statutory minimum wages in all sectors other than agriculture and domestic workers, streamlined the procedures for hiring and firing workers, and generally reduced the regulatory burden. Under the new system, manufacturing wages are set through a process of industry wide collective bargaining between workers and firms, although this process is still monitored by the government, and wage increases must be approved by the Ministry of Labor.

Although the collective bargaining arrangement applies, in principle, to all manufacturing firms, actual practice differs according to the firm's union status. Firms within any industry which have unionized labor forces do their collective bargaining in the context of Employment Councils, comprised of elected representatives of unions and employer organizations within that industry. Employment Council agreements have the force of law, although they must be approved by the Registrar for the Ministry of Labor. In non-union firms, the collective bargaining is conducted through Employment Boards, whose members are appointed by the Minister of Labor to represent employers and employees in a given industry, and whose agreements are non-binding recommendations to the Minister. In the future, the Employment Councils and Boards will be responsible for a wide range of labor relations issues including working conditions, individual dismissals for cause, and layoffs and retrenchments.

In this context, discussions of the role of unionization and minimum wages in driving the employer size wage differential have a different interpretation than is normally found in the literature. In Zimbabwe, no firms engage in firm-level collective bargaining with their own workers, but union firms participate in different collective bargaining institutions from non-union firms, so the presence of a union indicates a different institutional environment, and may generate different wage agreements. Under both institutional arrangements the collective bargaining agreement produces a minimum wage level for the coming year, so any individual firm is indeed bound by a minimum wage established by the Employment Council or Board of which it is a member and approved by the government. Differences in the impact of this minimum wage on firms' compensation decisions could produce the size effect.

4.2. Unionization

Unionization is measured by the employer's indication that at least some of its workers are members of labor unions. There is an extremely strong positive correlation between unionization and firm size, as seen in Table 7. If unions, or Employment Councils, have the effect of increasing wages, then the size effects may actually be picking up the higher level of unionization of larger firms.

This strong correlation, however, also makes it difficult to disentangle the effects of unionization and firm size. This is particularly true for the very large firms, all of which are unionized. Table 8 reports the results of including unionization dummies in both specifications. Again, columns 1 and 3 provide

Table 7
Unionization, minimum wages and ownership structure by firm size (in percent)

	All	Small	Medium	Large	Very Large
Workers who are in unionized firms	78.6	8.7	69.9	91.2	100.0
Workers in firms where minimum wage is binding	18.5	19.4	25.7	14.9	12.9
Mean share of foreign ownership (standard deviation)	13.7 (30.3)	4.67 (18.55)	6.4 (22.9)	22.9 (38.3)	16.5 (29.9)
Workers in parastatal firms	0.6	0	0	0	2.0
Workers in firms with some state ownership	6.0	0	0	8.0	13.0
Workers in cooperatives	4.5	13.0	8.0	0	2.0
Workers in firms with Zimbabwean ownersof European descent	45.0	10.0	44.0	50.0	53.0

Table 8
Earnings regressions including union dummies, dependent variable is log hourly earnings, absolute value of t-statistics in parentheses, $N = 1609$

	(1)	(2)	(3)	(4)
Ln(Size)	0.159 (8.83)	–	0.159 (7.21)	–
Medium	–	0.329 (3.73)	–	0.294 (3.08)
Large	–	0.484 (5.56)	–	0.437 (3.98)
Very Large	–	0.721 (7.36)	–	0.666 (5.52)
Union Dummy	–	–	0.001 (0.01)	0.067 (0.083)
Adjusted R^2	0.504	0.489	0.503	0.489

All regressions include controls for experience and its square, tenure schooling as in Table 3, training, race, gender and location.

reference estimates from Table 3. The union dummy has no effect on the magnitude of the firm size coefficient in the continuous specification, and its coefficient is insignificant. While the union coefficient is also insignificant in the discrete firm size specification, the size coefficients are slightly reduced. This could reflect the existence of size effects within the discrete size categories that are being captured by the union effect. [14]

It may also be the case that union effects vary by firm size, particularly if the bargaining power of unions or the level of government support for the union position varies with firm size. Table 9 presents estimates that allow size-variant unionization effects. Both specifications indicate some declining effect of unions

Table 9
Earnings regressions including interactive union effects, dependent variable is log hourly earnings, absolute value of t-statistics in parentheses, $N = 1609$

	(1)	(2)	(3)
Ln(Size)	0.203 (4.83)	–	–
Medium	–	0.330 (3.75)	0.252 (2.33)
Large and Very Large	–	0.594 (6.83)	0.627 (3.97)
Union	0.191 (1.04)	–	–
Union * Ln(Size)	−0.055 (1.14)	–	–
Small * Union	–	–	0.068 (0.50)
Medium * Union	–	–	0.131 (1.31)
(Large and Very Large) * Union	–	–	−0.016 (0.12)
Adjusted R^2	0.504	0.480	0.480

All regressions include controls for experience and its square, tenure schooling as in Table 3, training, race, gender and location.

[14] Although few white collar workers are themselves union members, the presence of a union in the firm has a similar effect on wages and estimated size coefficients for white and blue collar workers. Employment Board and Council agreements may set wages for some non-managerial white collar employees.

Table 10
Earnings regressions including union effects, small and medium firms only, dependent variable is log hourly earnings, absolute value of t-statistics in parentheses, $N = 705$

	(1)	(2)	(3)	(4)	(5)	(6)
Ln(Size)	0.183 (4.65)	0.169 (3.84)	0.183 (3.10)	–	–	–
Medium	–	–	–	0.267 (3.03)	0.198 (2.06)	0.193 (1.93)
Union	–	0.054 (0.59)	0.166 (0.55)	–	0.154 (1.66)	–
Union * Ln(Size)	–	–	−0.033 (0.35)	–	–	–
Small * Union	–	–	–	–	–	0.075 (0.61)
Medium * Union	–	–	–	–	–	0.140 (1.38)
Adjusted R^2	0.497	0.520	0.520	0.497	0.503	0.501

All regressions include controls for experience and its square, tenure schooling as in Table 3, training, race, gender and location.

at the larger firm sizes, and the discrete specification implies that unionization does generate some of the size effect in the medium size class, but this effect is not large. [15] If we restrict our attention to the 1265 workers in union firms, the coefficient on ln(size) is 0.147 with a standard error of 0.025, which is slightly smaller than that estimated for the whole sample. The coefficients (and standard errors) in the discrete specification are 0.206 (0.091), 0.312 (0.079) and 0.560 (0.092) respectively. Clearly, unionization is not the only cause of the size effect.

Another alternative is to use only the workers in small and medium firms, where there is more variation in unionization patterns. The results of this exercise are presented in Table 10. Unionization does appear to have some positive effect on wages, and to reduce size coefficients.

On the basis of the sum of this evidence, we cannot reject the hypothesis that some of the measured size effect is indeed due to greater unionization, and hence participation in Employment Councils, as firm size rises, and that this is particularly relevant in the smaller end of the size distribution. It is also clear that these institutional differences are not an exhaustive explanation of the size effect.

4.3. Minimum wages

Minimum wage laws, or minimum wages resulting from collective bargaining, have an effect on wages for those firms covered by the law only if those firms are in compliance, and then only if the level of the minimum wage is binding. If minimum wage compliance is greater in larger firms, or if the wage is binding in larger firms but not in smaller ones, minimum wages could generate the employer size wage premium.

[15] Since all very large firms are unionized, the two largest size classes have been combined in the discrete specification, and reference measures for this specification are presented in column (2) of Table 9.

Although the data do not include direct measures of either compliance or the relationship of the minimum wage to the firm's equilibrium wage, one question posed in the survey may provide some insight. Employers were asked whether a slight reduction in the minimum wage would change their hiring decisions by causing them to hire more workers. Those employers who answered that they would hire more workers can be considered to face a binding minimum wage constraint. Employers who responded in the negative are either not in compliance or are choosing wages at or above the minimum for other reasons. The share of workers in each size class whose employer reported that the minimum wage was binding is reported in Table 7. The share increases as we move from the small to medium size class, perhaps indicating more compliance among medium firms, and then falls sharply for the larger size classes. It is likely that the largest firms are in compliance, but that the level of the minimum wage is not binding.

This pattern makes it highly unlikely that the minimum wage is at the source of the size wage premium. Because the degree to which the minimum wage is binding is obviously a function of the level of wages being paid, this variable is clearly endogenous with respect to the wage. It cannot, therefore, be meaningfully included in earnings functions without introducing bias to the estimates of the size coefficient. The evidence in Table 7, however, suggests that while the minimum wage may generate some of the increase in earnings associated with working for a medium rather than a small firm, it is probably not responsible for the existence of the premium over the rest of the size spectrum.

4.4. Ownership structure, government relations and political economy

If larger firms are owned by different types of organizations or individuals than are smaller firms, and if these owners have motives for paying higher wages, then this would generate the size effect. We shall consider four types of owners who may have reason to pay increased wages, and explore the hypothesis that differences in ownership structure drive the size wage premium.

First, the sample includes one parastatal corporation with 2766 workers, and 10 other enterprises with some state ownership. If, for example, the parastatal enterprise is not operating on a profit-maximization principle, or is constrained to follow civil service wage guidelines, this could cause wages in this enterprise to be elevated. Similarly, the state, through its participation in joint ventures with the private sector, may require these firms to raise wages. Table 7 includes data on the number of workers in each size class working for firms with some state ownership.

The sample also includes 14 firms identifying themselves as cooperatives. In some of these cases, the workers are members of the cooperative, in others they are not. The cooperatives are concentrated in the smaller end of the size spectrum (see Table 7), and to the extent that workers in these firms are accepting lower wages in order to invest in the capital stock of the cooperative, this may contribute to the pattern of lower wages in smaller firms.

Table 11
Earnings regressions including ownership controls, dependent variable is log hourly earnings, absolute value of t-statistics in parentheses, $N = 1609$

	(1)	(2)	(3)	(4)
Ln(Size)	0.159 (8.83)	0.138 (7.26)	–	–
Medium	–	–	0.329 (3.73)	0.323 (3.71)
Large	–	–	0.484 (5.56)	0.416 (4.74)
Very Large	–	–	0.721 (7.35)	0.664 (6.85)
Parastatal	–	0.600 (6.13)	–	0.801 (9.81)
Some State Ownership	–	0.023 (0.26)	–	0.063 (0.64)
Cooperative	–	0.079 (0.56)	–	0.125 (0.81)
Percent of Foreign Ownership	–	0.004 (3.37)	–	0.004 (4.16)
Owner is Zimbabwean of European Descent	–	0.059 (1.21)	–	0.057 (1.51)
Adjusted R^2	0.504	0.518	0.489	0.513

All regressions include controls for experience and its square, tenure, schooling as in Table 3, training, race, gender and location.

Third, owners may use good labor relations to maintain good government relations. This may be important if the firm wishes to continue to receive benefits from government, or if the firm wishes to avoid difficulties with the government. Foreign-owned firms, for example, may pay higher wages to avoid accusations of worker exploitation, and hence be permitted to repatriate profits and prevent nationalization and expropriation. Multinationals may also be constrained to pay wages in keeping with corporate norms, particularly for white collar workers, and hence determine pay with reference to wages paid abroad. The share of foreign ownership is indeed greater in larger firms, as seen in Table 7. Similarly, Zimbabwean owners of European descent may also wish to avoid trouble with the government, and pay higher wages as part of this process. Firms owned by white Zimbabweans are also found in the larger end of the size spectrum, as indicated in Table 7.

A set of dummy variables measuring these aspects of ownership structure, as well as a continuous variable measuring the share of foreign ownership, have been included in the earnings regressions presented in Table 11. The effects generally have predicted signs, and are jointly significant at the 10% level. The size coefficients are reduced by the inclusion of these controls, indicating that ownership structure has some effect, but the size premium still persists.

If we include all of the controls implied by all of the hypotheses we have tested in a final earnings regression, we obtain some measure of the magnitude of the size premium left to explain. In the continuous size specification, the resulting coefficient on ln(size) is 0.103 (standard error 0.023), implying that a distance of one standard deviation above and below the mean of ln(size) generates a 32% wage premium. In the discrete size specification, these coefficients (standard errors) are: medium 0.234 (0.102), large 0.269 (0.116) and very large 0.479 (0.120), implying premiums of 26%, 31% and 61% respectively.

A similar exercise for the two occupational groups separately indicates another aspect of the size relationship that needs to be addressed. In the continuous size specifications, the coefficients (standard errors) generated in separate regressions including all the controls discussed above are 0.242 (0.034) for white collar workers, and 0.049 (0.020) for blue collar workers. This implies a two standard deviation size premium of 72% for the white collar workers compared with only 15% for their blue collar counterparts. Any explanation of the size premium then, must be consistent with much larger effects for white collar and highly skilled workers than for production workers, and the exploration of the efficiency wage explanations will take this issue into account.

5. Efficiency wage theories

The labor economics literature includes a large number of models which share the characteristic that under certain circumstances, raising wages may also increase firm profits. These 'efficiency' wage stories are generally based on the existence of relationships between wages and such factors as worker effort, worker turnover, esprit de corps, and vacancy. These models, developed to explain involuntary unemployment, can be used to explain firm size wage differentials if these potential relationships between wages, productivity and profits vary with firm size. Because the basic relationships between wages and effort or wages and applicant quality cannot be observed, direct tests of these hypotheses are generally not possible. Instead, this analysis will explore some of the implications of these hypotheses and attempt to assess their validity in the Zimbabwean manufacturing case. We will be particularly interested in arguments which might also explain why the premium received by blue collar workers is smaller than that received by white collar workers.

5.1. Efficiency wages and the hiring process

The first set of explanations focus on the hiring process, and involve the premise that raising wages increases either the size or the size and average quality of the applicant pool. In two related papers, Montgomery (1991) and Lang (1991) present models of the recruiting process in which some firms have an incentive to raise wages in order to increase the number of applications they receive per vacancy, and therefore reduce the probability that the vacancy will go unfilled. The extent to which any individual firm chooses to undertake this practice depends on the cost to the firm of an unfilled vacancy and the overall tightness of the labor market. In order for this approach to be a useful means of explaining the size premiums, large firms must have higher costs of vacancy, or be hiring for occupations with tighter labor markets.

Differences in the cost of vacancies can be generated by differences in capital intensity or in profitability per worker. If vacancies result in the idling of more capital or the failure to supervise more workers or a lack of accounting for larger amounts of business in larger firms, then these firms might choose to offer higher wages to avoid these vacancies. Capital intensity, as measured by the replacement value of plant and equipment per worker, does in fact increase with firm size in these data, ranging from a mean of Zim$25,000 in small firms to Zim$112,000 in very large firms.

The explanation also relies on the existence of tightness in the labor market. This does not seem plausible for blue collar workers, given the level of unemployment and underemployment in urban Zimbabwe. One does not have the impression that any firm need fear a vacancy on the factory floor will long go unfilled. The theory does seem more plausible for white collar and skilled technical personnel, who may be in short supply, and for whom the size premium is substantially larger. These workers make up a tiny fraction of the labor force of the small firms, so this argument is not useful in explaining the premium accorded by medium relative to small firms, but may explain the continued increase in wages offered by large and very large firms, and is consistent with the pattern of the premium across occupations.

The second argument focused on the hiring process is one put forth by Weiss (1980) and Weiss and Landau (1984). This formulation argues that by raising wages, firms can increase the average quality of their applicant pool by attracting a larger number of better quality applicants. The important dimension of quality in these arguments is one which cannot be easily observed by the firm (hence it must be something other than experience or schooling), but which is known to the workers, whose reservation wage is an increasing function of this characteristic. Alternatively, this attribute may be easily discernible by the firm, but not by other workers. If the firm is constrained by union rules or morale concerns to pay all similar workers (in terms of obvious attributes like schooling or experience) the same wage, then it will be concerned about the average quality of the applicant pool since it cannot discriminate by ability in setting wages once workers are hired.

This model can generate a relationship between wages and firm size in one of three ways. First, suppose that smaller firms have lower quality requirements than do larger firms. If none of the firms can accurately observe quality, this would result in larger firms, which seek a higher average quality pool, paying higher wages to attract these workers. If this practice works, we would expect workers in larger firms to be of higher ability on average, along a dimension that is unobservable to both the firm and the researcher. There is no way to test this hypothesis using these data.

A second possibility is that smaller firms are more able to pay individual workers based on their actual ability, and hence offer different pay to seemingly similar workers. This would result in a size premium only if small firms also had

Table 12
Hiring mechanisms as reported by firms and workers, distribution of responses in percent

	All	Small	Medium	Large	Very Large
Employer Responses					
Number of observations	198	38	65	47	48
Relative or friend of owner	8.1	26.3	9.2	0.0	0.0
Relative or friend of employee	19.7	21.0	24.6	14.9	16.7
Suggestion of business associate	0.5	0.0	0.0	0.0	2.1
Subtotal	28.3	47.3	33.8	14.9	18.8
Labor Office/Employment Agency	16.2	7.9	7.8	27.7	22.9
Trade or Technical School	1.0	0.0	1.5	0.0	2.1
Word of mouth	28.8	15.8	33.8	36.2	25.0
Formal Advertising	7.6	10.5	4.6	4.3	12.5
Hired from Factory Gate	14.1	10.5	15.4	12.7	16.7
Other	4.0	7.9	3.1	4.3	2.1
Worker Responses					
Number of observations	1609	149	556	456	448
Relative or friend of owner	11.7	42.3	13.7	6.6	4.2
Relative or friend of employee	28.3	23.1	28.1	33.4	25.2
Suggestion of business associate	1.6	3.4	1.4	1.3	1.5
Subtotal	41.6	68.5	43.2	41.2	31.0
Labor Office/Employment Agency	6.2	0.7	5.4	5.0	10.0
Trade or Technical School	0.9	0.0	0.0	2.2	1.1
Word of mouth	8.2	4.7	13.5	5.7	5.6
Formal Advertising	8.9	3.4	4.5	10.3	14.5
Went door to door	28.9	17.0	28.6	30.7	31.5
Other	5.2	4.7	4.8	4.6	6.2

lower quality standards. Otherwise, small firms would hire only the most able workers and pay them the highest wages. The ability of small firms to offer different pay to seemingly similar workers depends upon the balance of two factors. On the one hand, the small employer is likely to spend more time working with his staff and hence to be better able to actually identify differences in workers' quality. On the other hand, morale concerns are likely to be equally important in small as in large firms, particularly if workers in small firms value a familial atmosphere. It is also more likely that workers in smaller firms will know one another's wages, making the potential morale consequences even more severe.

The third possibility is that firms of different sizes have different screening costs, and hence choose different combinations of applicant screening and wage offers to obtain the quality of worker they prefer. Although screening costs cannot be measured, we do have information about the mechanisms firms use to hire workers, and about the way individual workers found their jobs. Employers were asked "What is the common way of finding workers?" The distribution of answers across the 198 firms who responded are presented in Table 12. "Workers were his job?'' The distribution of these answers is also presented in Table 12. 'I went door to door' from the workers' responses is equivalent to a firm response of 'hiring from the factory gate'.

The responses in the table are divided into two groups. The first group, which includes hiring one's own friends and relatives, or the relatives and friends of current employees or business associates, involves selecting workers from among people with whom one has a personal relationship. The other mechanisms involve selecting workers from a pool of strangers. Using the more personal hiring mechanisms implies that the firm may have substantial information about each applicant, who is known to them or can be assumed to share traits with a current employee. The more impersonal hiring mechanisms would require some screening to obtain that information. If this screening is costly, firms may find it profitable to use a high wage offer to obtain an applicant pool of higher average quality and reduce their need for expensive screening activities. If larger firms are more likely to use impersonal techniques this could generate higher wage offers from larger firms.

The data in Table 12 indicate that larger firms do indeed use more impersonal hiring strategies than do smaller ones. One explanation for this choice could be that larger firms, which make many more hires of a larger variety of types of workers, are less able to use personal connections and networks to fill all their vacancies. Whatever the explanation, the evidence is consistent with a strategy in which smaller firms hire people they know, or have relatively easy access to information about, while larger firms hire strangers and use higher wages to increase the quality of the applicant pool. The choice of a specific hiring mechanism is clearly endogenous, with firms choosing their hiring strategy and their wage offer simultaneously. It is, therefore, inappropriate to include controls for hiring mechanism in the earnings function, so we cannot examine the effect of including such controls on the size coefficient, though it seems clear that differences in this behavior are correlated with firm size.

5.2. *Efficiency wages and shirking*

The second class of efficiency wage arguments are those associated with worker behavior on the job. The most familiar of these arguments is that firms increase wages as a mechanism to decrease shirking. The expected cost of shirking to a worker is the probability of being caught (and therefore fired) multiplied by the cost of job loss. Firms can obviously increase this cost either by raising the probability of catching the worker through increased monitoring, or by increasing the cost of job loss by raising wages relative to the workers' next best offer.

A shirking argument could create a size premium in two ways. If the cost of shirking to the firm is the same for all firm sizes, but monitoring is more expensive for larger firms, we would expect to see larger firms paying higher wages and engaging in less monitoring than smaller ones. [16] The larger physical size, more complex production processes and weaker personal ties between workers and management that characterize larger firms could raise monitoring costs in these firms. If monitoring of white collar workers is more difficult, this would also generate the larger premium accorded to white collar workers. In this case, we would expect to see an inverse relationship between monitoring inputs and wages, and hence between monitoring inputs and firm size.

It is also possible that the cost of shirking to the firm is higher in larger firms. Large firms are more likely to use assembly line production processes in which low output by one worker reduces output at all subsequent points on the line. The greater capital intensity of larger firms, and the more sophisticated equipment they use, implies that the potential damage generated by inattentive or negligent workers is much greater than in smaller firms. In this case, we would expect larger firms to invest more resources in deterring shirking, and hence both to pay higher wages and use more monitoring inputs. If both arguments hold, we would find larger firms paying higher wages, but there is no clear prediction about the relationship between monitoring inputs and firm size.

Because firms choose their wage offers and their monitoring inputs simultaneously, it would be inappropriate to attempt to examine the usefulness of shirking models by including monitoring inputs in the earnings function. We may, however, gain some insight into the relevance of this explanation through an examination of the relationship between monitoring inputs and firm size. Monitoring inputs can be measured using data about labor force structure. The ratio of managerial workers to the total number of workers in the firm is a broad measure of monitoring activity. The monitoring of production workers is done by supervisors and foremen in larger firms and by the owner or manager in smaller ones. The ratio of the number of supervisors to the number of production workers provides a proxy for monitoring inputs per production worker. For firms in which those tasks are performed by managers or owners, a similar ratio can be calculated using several different assumptions about the fraction of the time the manager spends in supervising production. Mean values of these ratios for the firms in the sample are presented in Table 13.

The data in the table show, first, that although only five of the small firms employ supervisors, under most assumptions about the time managers spend supervising production, these firms have the potential to monitor their production workers much more closely than do firms in the other size categories. [17] It seems reasonable, then, to conclude that at least part of the wage difference between

[16] It is also possible that monitoring costs are lower in large firms, which would be able to employ workers who specialize in monitoring. Such a circumstance would not, however, generate a positive size premium.

[17] Nineteen of the medium sized firms also employ no supervisors. All of these have 35 or fewer workers, and the foregoing characterization of small firms would be relevant to them as well.

Table 13
Monitoring inputs by firm size, mean values by size class

	All	Small	Medium	Large	Very Large
Number of Firms	201	40	66	47	48
Mean value of ratio of managerial workers to firm size	0.08	0.234	0.064	0.027	0.024
Number of firms employing supervisors	147	5	47	47	48
Mean value of ratio of supervisors to production workers for firms employing supervisors	0.083	0.446	0.083	0.072	0.057
Mean value of ratio of supervisory labor to production workers assuming managers in firms without supervisors spend 10% of their time supervisingproduction	0.067	0.084	0.062	0.072	0.057
Mean value of ratio of supervisory labor to production workers assuming managers in firms without supervisors speend 25% of their time supeervising production	0.076	0.124	0.066	0.072	0.057
Mean value of ratio of supervisory labor to production workers assuming managers in firms without supervisors spend 50% of their time supervising production	0.092	0.192	0.073	0.072	0.057

small firms and the others could be driven by these differences in the ease of monitoring. Among the firms in the larger size categories, there is a weak negative correlation between firm size and monitoring inputs, measured by any of the proxies in Table 13. This is consistent with the idea that monitoring is more expensive in larger firms, though the differences in monitoring inputs across firm size do not seem large enough to explain the substantial wage differentials. In general, these data are not inconsistent with a shirking based approach.

The shirking argument also requires that a worker caught shirking can indeed be dismissed. Labor legislation in Zimbabwe, however, makes it extremely difficult to fire a worker classified as permanent by having been continuously employed for three months. Although recent changes in labor legislation have made dismissal easier, it still requires bureaucratic approval. Employers respond by maintaining a stock of casual employees under fixed term (but often renewed) contracts of less than three months. The use of such casual workers increases with firm size, with 35% of small firms using them as compared with 83% of very large firms. Most firms using casuals (68%) indicated that these workers generally received the same wages, allowances and benefits as permanent workers, being distinguished only by the fact that they can be more easily fired. Many firms may use casual contracts as a labor screening device, which is particularly important if workers cannot be fired once they become permanent. This evidence indicates that firms view these firing restrictions as binding and likely to be enforced.

The use of casuals, however, does not imply that firms are able to avoid the dismissal regulations entirely. Among those firms employing them, casual workers comprise an average of 21% of the firms' labor force, and are almost exclusively emplo Since firms do not choose to have all of their employees as casuals, they face real limitations on their ability to fire workers, even for cause. [18] This is particularly true for white collar workers, almost none of whom are employed as casuals. These institutional constraints, which are common in sub-Saharan Africa, diminish the potential importance of the shirking model as a fundamental cause of the size premium.

5.3. *Efficiency wages and voluntary turnover*

A more plausible argument is that firms use higher wages as a means of reducing voluntary turnover, as modelled by Salop (1979). If maintenance of a stable labor force is more important for larger firms because of higher training costs, larger investments in both specific and general human capital, a greater need for teamwork and coordination or greater recruiting costs, then these firms may find it profitable to raise wages to deter quits. As we saw in Table 6, there is indeed a strong negative correlation between quit rates and firm size. Workers in larger firms are also more likely to be receiving on the job or other training and, as we have seen, the screening costs associated with recruitment may also be greater in larger firms. Greater capital intensity, which is likely to be associated with the use of more specialized equipment, may increase basic training costs as well. These factors indicate that larger firms may indeed have greater need for labor force stability, and hence offer wages above market clearing levels in order to generate it.

If stability is more important in white collar than in blue collar staff, this explanation would also be consistent with the different magnitudes of the size premiums across occupations. If, for example, the training any firm provides its white collar staff, such as accounts clerks, contains a large component of general skill development, there will be a high risk of worker turnover. Since the identification and screening of job candidates and the provision of training are likely to be quite costly, raising wages to retain such workers may well be a profit maximizing strategy for large firms. Anecdotal evidence gathered during the survey indicated that many firms consider 'poaching' of their white collar staff by competitors to be a significant problem.

Clearly, the fact that larger firms have lower turnover could be an outcome, rather than a cause, of the fact that such firms pay higher wages. There is some other evidence, however, that larger firms' wage structures are generally conducive to promoting longer tenure on the job. First, workers in larger firms receive

[18] There are a number of reasons why firms would choose to have some permanent workers despite the fact of dismissal restrictions, including the need to preserve firm-specific human capital by reducing turnover.

higher returns to experience. The survey asks workers what their earnings were when they first came to work for the firm, which allows calculation of the average annual rate of wage growth for each worker. Controlling for all the aspects of worker quality generally included in earnings functions, we find that workers in larger firms experienced more rapid average wage growth. This result is echoed in earnings functions estimates which allow the returns to experience to vary by firm size, which also show higher returns to experience in larger firms. These wage profiles are also consistent with shirking-based efficiency wage models, though the institutional constraints on firing discussed earlier make the turnover explanation appear more plausible.

Second, workers in larger firms are also more likely to have been promoted, in the sense of changing job classifications, over the course of their career, which is an additional manifestation of the use of advancement as a means of retaining good workers. Only 11.4% of the workers in small firms had changed jobs, compared with 41.5% of the workers in very large firms. Although this evidence does not constitute a direct test of the turnover hypothesis, it does provide strong support for this explanation of the size effect.

5.4. *'Sociological' arguments, gift exchanges and rent sharing*

Akerlof (1982, 1984) argues that employers may pay wages above the minimum required to hire workers as a gift, in response to workers' gifts of effort above the minimum required by the job. Although neither the entire wage payment nor the entire effort of the worker can be construed as gifts, part of each may be a gift exchange. Workers' effort norms are conditioned by their treatment by the firm, in terms of remuneration and work rules. The payment of gifts induces better worker performance by creating an environment in which workers believe the extra effort is appropriate.

As Akerlof points out, the idea that labor contracts contain a gift exchange component is related to the notion of 'fair' wages. Workers who feel that their wage payment is fair relative to some standard (and the nature of that standard is itself an interesting question) will feel more loyal to their employers and be more willing to engage in behavior, such as extra effort, that ultimately benefits their employers. By allowing workers to feel good about working harder, higher and more 'fair' wages eliminate some of the incentive incompatibility inherent in labor contracts.

The notion of a 'fair' wage does not come solely from the workers' perspective. Although the questionnaire used to gather these data did not ask firms to describe their wage setting practices, several of the people interviewed by the author were willing to discuss how their salary scales were determined. Notions of fairness were common in the discussions, and fairness took a number of dimensions, including concerns about the appropriate relationship between managerial and blue collar wages, notions of a 'living' wage on which an individual could raise and educate children, comparisons of wages in the firm with wages elsewhere in the economy, and comparisons of wages in the firm to the profits earned.

Gift exchange or fairness notions may generate a size effect in two ways. If larger firms prefer the use of high morale to the use of monitoring as a means of generating and maintaining worker productivity, they would need to provide larger gifts to workers to induce larger gifts of effort from workers. Alternatively, the acceptable gift or fair wage required to produce any level of effort may be greater in larger firms. Akerlof's models are intended to explain involuntary unemployment, and hence provide no discussion of what might generate variation in the size of gifts or in the level of wages deemed to be fair. One possibility is that the magnitude of an acceptable gift is a function of the resources available to the giver. If larger firms are 'able' to grant larger gifts, perhaps by virtue of earning larger profits, workers may require larger gifts from them in order for the gift exchange to involve equal 'sacrifices' on both parts. Similarly, a notion of what constitutes a fair wage could also be a function of workers' perception of firm performance, so that workers expect to earn more when the firm is doing well.

These stories lead to a rent sharing notion of wage determination, in which rent sharing is simply one form of the efficiency wage argument. In order for rent sharing to explain the size premium, rents must be correlated with firm size. The appropriate measure for rent would be economic profit, but the data do not permit this to be calculated. The data do allow the calculation of several correlates or proxies of rents: sales per worker, value added per worker (sales minus raw materials costs, rent, advertising expenditures and utilities), and accounting profits per worker (value added minus the wage bill). Variation in these measures could, in principle, be generated entirely by variation in the capital-labor ratio across firms, so that accounting profits would vary even if all firms had zero economic profit. In these data, however, the correlation between capital–labor ratios and these proxies for rents is small, indicating that these measures are indeed also correlated with economic profit or rents. Although all of these measures are strongly correlated with each other, none is positively correlated with firm size. Profits per worker are clearly a function of wages, and therefore endogenous with respect to the wage decision, but the other two measures are not. Table 14 presents the results of including each of the rents proxies in the earnings function. [19] While there is a positive coefficient on all of the rents proxies, indicating that rent sharing is a component of the wage setting process, the size coefficients are actually larger in the specifications including the rent proxies. Rent sharing does appear to be important, and may well result from a gift exchange or fairness motive, but it does not explain the size effect.

[19] Missing values for some components of profits reduces our sample to 1573, and baseline size coefficients for this sample are presented in columns 1 and 5.

Table 14
Earnings functions including controls for profits, dependent variable is log hourly earnings, absolute value of t-statistics in parentheses, $N = 1573$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ln(Size)	0.109 (4.934)	0.113 (5.135)	0.112 (5.043)	0.115 (5.637)	–	–	–	–
Medium	–	–	–	–	0.257 (2.632)	0.261 (2.607)	0.265 (2.674)	0.230 (20.342)
Large	–	–	–	–	0.274 (2.414)	0.290 (2.530)	0.283 (2.480)	0.253 (20.226)
Very Large	–	–	–	–	0.509 (4.385)	0.526 (4.513)	0.517 (4.438)	0.484 (40.250)
Profits per Worker	–	5.04×10^{-7} (2.827)	–	–	–	4.95×10^{-7} (2.740)	–	–
Value Added per Worker	–	–	1.39×10^{-7} (0.900)	–	–	–	1.18×10^{-7} (0.730)	–
Sales per Worker	–	–	–	1.21×10^{-6} (2.667)	–	–	–	1.08×10^{-6} (20.386)
Adjusted R^2	0.688	0.692	0.689	0.698	0.688	0.692	0.697	0.696

The regressions include all the controls discussed in Section 2 and Section 3.

If the impersonal environment in larger firms creates greater feelings of alienation or hostility on the part of workers, they may need larger gifts or higher compensation in order to happily provide more effort. If this were the case, however, one would expect to see larger premiums paid in the blue collar occupations, where work is more tedious, work rules are more rigid, and workers are more likely to feel distanced from management and without control of their work environment.

Finally, notions of internal fairness may also be important. Turnover considerations may be the source of size premiums for white collar workers, and internal equity rules may generate size premiums for the rest of the staff. If the firm feels that equity is important, it may also raise wages for blue collar production and support staff workers as well, though that premium would be smaller. In this sense, 'a rising tide raises all boats'.

These sociological explanations could well lie at the heart of the size premium. Larger firms do differ from smaller ones in work norms and cultures, in the importance of personal relationships to the work environment, in their relations with government and other firms, and in the production processes they employ. These differences seem particularly stark in Zimbabwe, where smaller firms replicate family and village based production patterns and social networks, while larger firms are structured around a foreign and unfamiliar set of social norms. Direct tests of these explanations, like tests of the role of subjective working conditions, are not possible with these data.

6. Conclusions

This paper sought to answer two questions. First, what does the employer size wage effect tell us about the Zimbabwean labor market? The strongest conclusion one can reach is that the manufacturing labor market in Zimbabwe functions much like labor markets elsewhere. It too, is characterized by deviations from the law of one price, and the explanations for these deviations that seem plausible for Zimbabwe are also those that seemed plausible elsewhere. To the extent that the Zimbabwean market differs from others, it is a difference in degree, not a difference in kind. Government intervention is more extensive, and the information and moral hazard problems that generate efficiency wages may be more severe, but the fundamental functioning of this market seems to be the same as that seen in developed countries. This indicates that the development process may reduce these differentials, bringing the market closer to competition, though results from developed countries indicate that perfect competition may not be achieved.

Second, what can the case of Zimbabwe tell us about the employer size wage effect in general? Although government intervention in Zimbabwe has been much more extensive than it has been elsewhere, unionization, minimum wage and political economy explanations do no better at explaining the premium in Zimbabwe than they do elsewhere, supporting the notion that these effects are not generated primarily by institutional intervention. The use of matched employer–employee data allows us to explore some efficiency wage hypotheses in more detail, and generates the same basic conclusions as found elsewhere in the literature: turnover, hiring and sociological explanations seem more plausible than those based on shirking. If this is the case, then the solution to this mystery will require the gathering of more subjective data than has been available in the past.

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Appendix A. Variable Definitions

Hourly Earnings	Weekly Wages + Weekly Allowances/Usual Weekly Hours
Experience	Number of years the individual has been out of school
Tenure	Years of employment with current employer
Primary	= 1 if highest level of schooling completed is a primary grade
Some Secondary	= 1 if highest level of schooling completed is a secondary grade short of complete
Complete Secondary	= 1 if highest level of schooling completed is the final year of secondary school
University	= 1 if highest level of schooling completed is at university
Polytech	= 1 if individual attended polytechnical school
Vocational	= 1 if individual attended vocation school
Former Apprentice	= 1 if individual was an apprentice, in this firm or elsewhere
On the job training	= 1 if individual is currently receiving on the job training
Outside training	= 1 if individual is currently receiving training outside the job
Male	= 1 if individual is male
European	= 1 if individual is of European descent

Asian	= 1 if individual is of Asian descent
Harare	= 1 if individual resides in Harare (the capital)
Bulawayo	= 1 if individual resides in Bulawayo
Ln(size)	Log of firm size
Medium	= 1 if firm size is between 11 and 100
Large	= 1 if firm size is between 100 and 250
Very Large	= 1 if firm size is greater than 250
Hours	Usual weekly hours
No Electricity	= 1 if the firm has no electricity
No Phone	= 1 if the firm has no telephone
Bad Transport	= 1 if the firm said that worker transport was a problem
Quit-rate	Number of quits in the last year/firm size
Union	= 1 if the any of the firm's workers are unionized
Parastatal	= 1 if the firm is wholly owned by the state
State-owned	= 1 if the firm has some state ownership
Cooperative	= 1 if the firm is a cooperative
Percent of Foreign Ownership	Fraction of firm's shares owned by foreigners of by foreign firms
Owner is Zimbabwean of European Descent	= 1 if firm is Zimbabwean owned and the majority if of European descent
Profits per worker	Accounting profits (sales-raw materials-utilities-rent-advertising expenditures-wages)/size
Value added per worker	(Sales - raw materials - utilities - rent - advertising)/size
Sales per workers	Sales/size

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